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# Risk of Kawasaki Disease/Multisystem Inflammatory Syndrome Following COVID-19 Vaccination in Korean Children: A Self-Controlled Case Series Study

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## ABSTRACT

**Background:** Rare cases of Kawasaki disease (KD) and multisystem inflammatory syndrome in children (MIS-C) have been reported following the coronavirus disease 2019 (COVID-19) vaccination; however, the association between COVID-19 vaccination and the risk of developing KD/MIS-C has not yet been established.
**Methods:** We conducted a self-controlled case series analysis using a large-linked database that connects the COVID-19 immunization registry with nationwide claims data. We identified individuals aged < 18 years who received their initial COVID-19 vaccination and had a KD/MIS-C diagnosis with a prescription for intravenous immunoglobulin or corticosteroids between October 18, 2021, and April 15, 2023. The observation period was set as 240 days from the date of the COVID-19 vaccination. The risk window was 60 days after vaccination, with the remaining observation period serving as the control window. The incidence rate ratios (IRRs) and 95% confidence intervals (CIs) in the risk versus control windows were estimated using the conditional Poisson regression model. We further analyzed the vaccine doses and types for secondary analysis. We also performed subgroup analyses stratified by sex, age, comorbidities, and other conditions and sensitivity analyses by varying the length of the risk window and outcome definition.
**Results:** Among 2,369,490 individuals who received the COVID-19 vaccination, 12 cases of KD/MIS-C were identified, which included five and seven patients in the risk and control windows, respectively. There was no increased risk of KD/MIS-C within the 60-day period of vaccination (IRR, 0.53; 95% CI, 0.17–1.60). Secondary subgroup and sensitivity analyses showed no significant increase in the risk of KD/MIS-C after COVID-19 vaccination, which is consistent with the results of the main analysis.
**Conclusion:** The results of this nationwide study suggest that the risk of developing KD/MIS-C did not increase after COVID-19 vaccination. However, owing to the lack of a sufficient number of cases, future studies utilizing multinational long-term follow-up databases should be

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**Disclosure**
The authors have no potential conflicts of interest to disclose.

**Author Contributions**
Conceptualization: Kim S, Ko HY, Oh J, Yoon D, Kim JH, Shin JY. Funding acquisition: Shin JY. Supervision: Choe YJ, Shin JY. Writing - original draft: Kim S, Ko HY. Writing - review & editing: Kim S, Ko HY.

conducted. Considering the increasing incidence of KD/MIS-C and the limited understanding of its precise biological mechanisms, additional research on KD/MIS-C is warranted.

**Keywords:** Kawasaki Disease; Multisystem Inflammatory Syndrome in Children; COVID-19 Vaccine; Self-Controlled Case Series

## INTRODUCTION

Coronavirus disease 2019 (COVID-19) vaccines have been validated to be effective in preventing severe illness or death caused by severe acute respiratory syndrome coronavirus-2 (SARS-CoV-2) infection in children.[1,2] The Centers for Disease Control and Prevention recommended all individuals aged 5 years and older receive a single dose of the updated COVID-19 vaccine (Pfizer-BioNTech, Moderna, or Novavax).[3] Infants and children aged 6 months to 4 years are recommended to receive the updated COVID-19 vaccine, starting with at least one initial dose, followed by subsequent doses as necessary to remain up to date. To date, COVID-19 vaccines have undergone rigorous safety analyses in the US, supported by extensive clinical trial data and vaccine safety monitoring systems, such as the Vaccine Adverse Event Reporting System (VAERS), Vaccine Safety Datalink, and the Clinical Immunization Safety Assessment Project. Nevertheless, rare but serious adverse events have been documented in vaccinated pediatric populations.[2,4] Therefore, evaluating the association between COVID-19 vaccine administration and the subsequent adverse effects is crucial.

Kawasaki disease (KD) is an acute multisystemic vasculitis primarily affecting children between 6 months and 5 years of age. It presents with symptoms such as high fever, cervical lymphadenopathy, and mucocutaneous inflammation.[5] The cause of KD remains uncertain; however, evidence suggests that preceding respiratory infections, such as those caused by human adenovirus and human coronavirus, likely play a role in triggering KD.[6,7] Multisystem inflammatory syndrome in children (MIS-C), a novel syndrome with clinical manifestations similar to KD, was first reported in the United Kingdom in April 2020,[8] and is characterized by symptoms including fever, rash, and multiple organ dysfunction following infection with SARS-CoV-2. Although MIS-C presents with clinical symptoms similar to those of KD, it differs in that the average age of onset is higher and encompasses a broader range of age group.[9]

Recent reports of MIS-C and KD occurring after COVID-19 vaccination have raised public health concerns.[10,11] However, several studies have evaluated the effectiveness of COVID-19 vaccination in reducing the occurrence of KD/MIS-C by preventing COVID-19 infection.[12-14] Reports from VAERS indicate that the rate of MIS-C occurring after COVID-19 vaccination appears to be rare, with only one report per million vaccinated individuals, resulting in a lack of studies describing a clear association.[11] Although the connection between COVID-19 vaccination and the onset of KD/MIS-C in children remains unclear, there is insufficient data to elucidate a clear relationship between these two events.

To address this knowledge gap, this study used a large-linked database from South Korea to evaluate the association between KD/MIS-C and COVID-19 vaccination using a self-controlled case series (SCCS) analysis.

# METHODS

## Data source

This study was conducted as a part of South Korea's COVID-19 Vaccine Safety Research Committee (CoVaSC). CoVaSC conducts comprehensive analyses of COVID-19 vaccine safety, leveraging the National Health Insurance System (NHIS) reimbursement data to investigate the causality of reported adverse events following COVID-19 vaccination, and provide accurate information to the public.

Data for the study were extracted from a large-linked database that integrated the Korea Disease Control and Prevention Agency's (KDCA) COVID-19 immunization registry (February 26, 2021 to April 15, 2023) with the NHIS claims data (January 1, 2002 to April 15, 2023). Since the initiation of the COVID-19 vaccination in South Korea (February 26, 2021), the KDCA has collected detailed vaccination data, such as the vaccination date, administered dosage, and vaccine type, for the entire population.[15] Approximately 97% of the Korean population is covered by the National Health Insurance Program, and the NHIS claims database contains details of the medical services that have been provided. Using the linkage between these two data sources, we extracted relevant information from vaccination and medical utilization records to construct the analysis dataset. Diagnostic records were coded according to the International Classification of Diseases 10th revision (ICD-10). Prescription drug information was identified through domestic drug chemical codes based on the Anatomical Therapeutic Chemical (ATC) classification of the World Health Organization.

## Study population

We identified individuals under 18 years of age who received their first COVID-19 vaccination between October 18, 2021, and April 15, 2023. Of these, we included those with a diagnostic record for KD/MIS-C who were prescribed intravenous immunoglobulin (IVIG) or corticosteroids (CS) following COVID-19 vaccination during this period. Individuals who received the COVID-19 vaccine outside South Korea, including foreigners and participants in clinical trials, were excluded from this study, as their post-vaccination healthcare utilization may not have been comprehensively documented. Additionally, to prevent potential misclassification of exposure, individuals with incomplete immunization records, such as missing detail on vaccine types and dose number, were excluded from the analysis. Moreover, individuals with a diagnostic record of KD/MIS-C in the 365 days preceding their initial COVID-19 vaccine dose were excluded to ensure focus on incident outcomes. Furthermore, individuals without a diagnostic record of KD/MIS-C within the pre-specified observation period (240 days after the first dose of the COVID-19 vaccine) were also excluded. The study population selection process is illustrated in **Fig. 1**.

## Study design: self-controlled case series analysis

We used SCCS analysis to assess whether the occurrence of KD/MIS-C is associated with COVID-19 vaccination. It was originally proposed to investigate potential associations between vaccines and adverse events involved by comparing the relative incidences of these events during a risk window in comparison to a control window.[16] This design was developed to address research questions where establishing an appropriate comparison group is difficult and has been widely used for vaccine safety assessment.[17-19] The SCCS analysis, which divides time periods within individuals without needing a separate control group, minimizes bias from time-invariant confounders. The SCCS analysis, however, relies on several assumptions, particularly that the occurrence of adverse events should

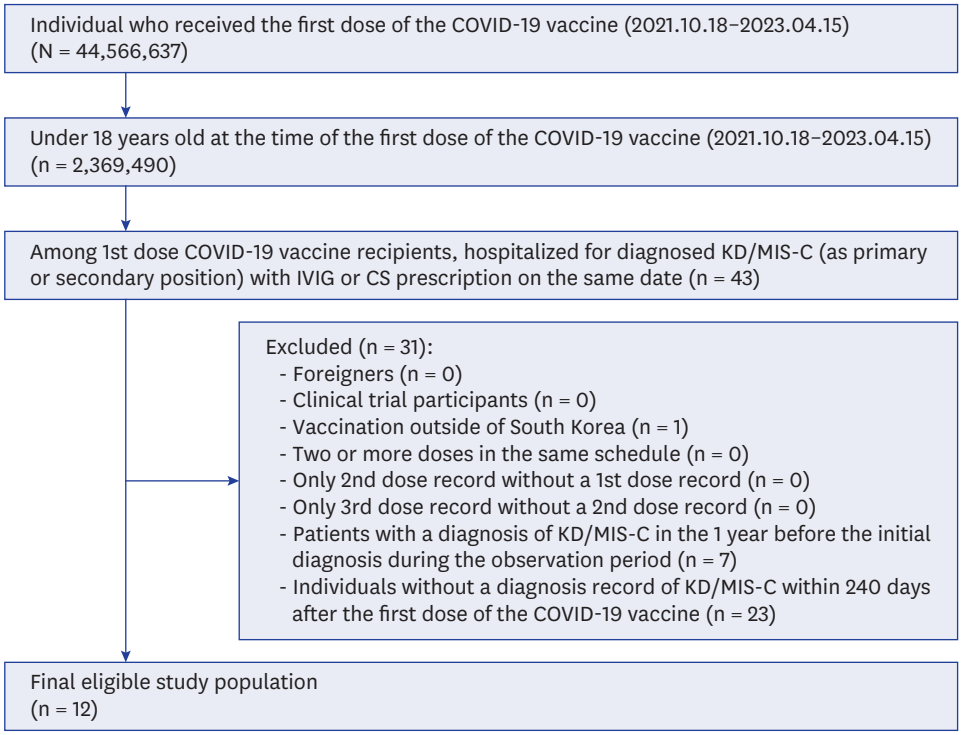


**Fig. 1.** Flow chart of subjected patients.
COVID-19 = coronavirus disease 2019, KD = Kawasaki disease, MIS-C = multisystem inflammatory syndrome in children, IVIG = intravenous immunoglobulin, CS = corticosteroids.

be independent and should not influence subsequent events. Consequently, only the first episode was considered the outcome of interest for this study. We prespecified the observation period as 240-days from the first COVID-19 vaccination date to minimize the impact of time-varying confounders. If the study end date (April 15, 2023) came before the end of the prespecified observation period, observations were limited to the study end date. The risk window was defined as 1–60 days after each dose,[13] and the control window was established as the period outside of the risk window (with the vaccination date considered as day 0) (**Fig. 2**). We defined 60 days of risk window since neutralizing antibodies against SARS-CoV-2 lasted for several months post vaccination in clinical trials for COVID-19 vaccines.[20,21] We also conducted sensitivity analyses varying the length of risk window.

## Exposure

In this study, exposure was defined as any COVID-19 vaccination administered within the observation period, which included up to four doses. The following vaccines were available during the study period: BNT162b2 (Pfizer-BioNTech) and NVX-CoV2372 (Novavax). Unlike mRNA technology-based BNT162b2, NVX-CoV2372 utilizes a genetic recombinant approach with synthetic antigens, similar to vaccines for Hepatitis B and cervical cancer, that have been extensively administered. NVX-CoV2372 was indicated exclusively for adolescents (12–17 years old).[22] Homologous vaccination was categorized for participants who consistently received the same COVID-19 vaccine type for all doses, while heterologous vaccination referred to those who received different vaccine types across any of the doses.

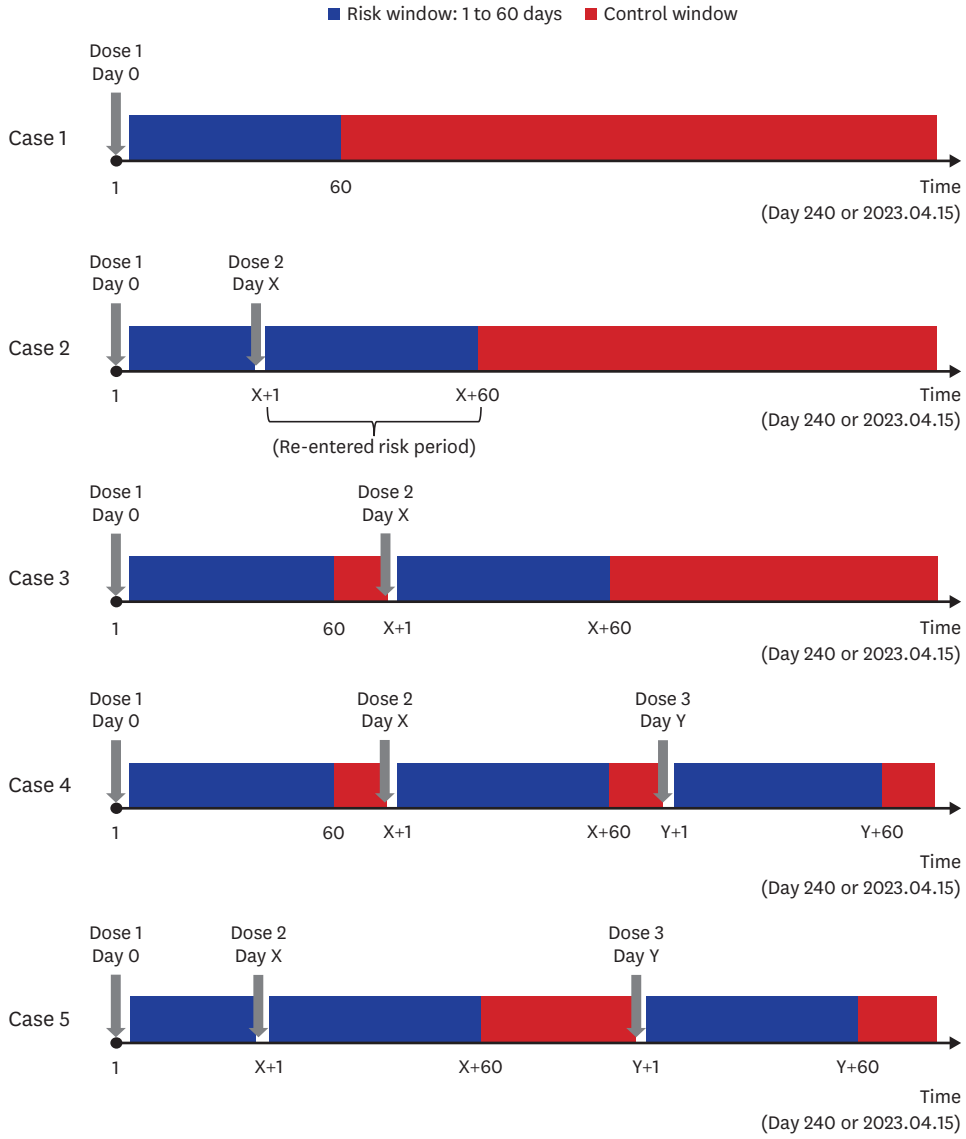


**Fig. 2.** Setting of risk window and control window in self-controlled case series model.

## Outcomes

The outcome of interest was the composite endpoint of KD and MIS-C after COVID-19 vaccination. These excessive inflammatory events were reported using the ICD-10 and specific ATC codes. To identify a definite diagnosis of MIS-C, individuals diagnosed with bacterial sepsis, toxic shock syndrome, infective myocarditis, or myocarditis were excluded. Additionally, we restricted the outcome to hospitalized patients with a primary or secondary diagnosis (major disease that caused patient to seek medical care) of KD/MIS-C, accompanied by a medication order for IVIG or CS on the same date, to enhance the validity of the outcome (**Supplementary Table 1**).

## Statistical analysis

Descriptive analyses were performed based on the study population to assess the baseline characteristics of patients who developed KD/MIS-C during the risk or control windows after COVID-19 vaccination. Continuous variables such as age were summarized by calculating

the mean with standard deviation and assessed using *t* tests for statistical significance. Categorical variables are summarized as frequencies and percentages, with statistical significance assessed by the $\chi^2$ test. Demographic characteristics (age, sex, health insurance type, and residence) were evaluated on the date of the first COVID-19 vaccination. To measure the Charlson comorbidity index (CCI) and comorbidities, including malignant neoplasms, type 1 diabetes mellitus, congenital malformations of the circulatory system, asthma, neurodevelopmental disorders, epilepsy, and depression/dysthymic disorder, we analyzed medical claims data from the year before vaccination.

A conditional Poisson regression model was used to compare the incidence rate ratios (IRRs) and 95% confidence intervals (CIs) for KD/MIS-C between the at-risk and control windows. In the secondary analysis, we analyzed outcomes by vaccine dose and type. Additionally, we performed subgroup analyses by dividing the groups according to characteristics that could influence the prevalence of KD/MIS-C. The subgroups were stratified by sex (male, female), age groups (0–4, 5–11, and 12–17), CCI (< 3, 3+), pre-vaccination comorbidities, socioeconomic status (health insurance subscriber/medical aid beneficiary), and residence (metropolitan or state).

We conducted a range of sensitivity analyses to evaluate the robustness of the final outcome and to understand how the various definitions and assumptions employed in the analysis affected the results. First, to analyze the effect of the length of the risk window on the study results, we repeated the analysis by varying the risk window from 1–30 days to 1–90 days post-vaccination. Second, to assess the reliability of the outcome definitions, we redefined the outcomes as follows: 1) having a diagnosis of KD/MIS-C and a prescription for IVIG administration; and 2) having a diagnosis of KD/MIS-C and a prescription for IVIG or other therapeutic agents, including aspirin, CS, anakinra, infliximab, tocilizumab, acetylsalicylic acid, enoxaparin, and warfarin. Third, to account for the impact of including susceptible population for the outcome occurrence, and to establish a meaningful study population for identifying associations between the vaccine and KD/MIS-C, we excluded the following individuals: 1) individuals who died within 7 days of developing KD/MIS-C; 2) individuals with COVID-19 in the 60 days prior to receiving the COVID-19 vaccine; 3) individuals with COVID-19 infection in the 60 days before the diagnosis of KD/MIS-C; 4) individuals diagnosed with KD/MIS-C on the same date as the diagnosis of bacterial sepsis, toxic shock syndrome, or infectious myocarditis. All statistical analyses were conducted using the SAS Enterprise Guide, version 8.3 (SAS Institute Inc., Cary, NC, USA).

### Ethics statement

This study was approved by the Public Institutional Review Board designated by Ministry of Health and Welfare and informed consent was waived (P01-202203-01-005).

## RESULTS

Between October 18, 2021, and April 15, 2023, a total of 2,369,490 individuals under the age of 18 who received the first dose of the COVID-19 vaccine were identified. Among these, 43 cases of KD/MIS-C were identified during the observation period. After applying the exclusion criteria, 12 patients were included in the SCCS analysis. All 12 cases of KD/MIS-C that occurred during the risk or control windows were within the age group of 12–17 years. During the risk window, the mean age of the five individuals with KD/MIS-C was 14.8 years, with 60.0% being

female. In contrast, during the control window, the mean age of the seven individuals was 15.1 years, with 28.6% being female. The CCI for the 12 KD/MIS-C cases during the risk and control windows was consistently < 3. The CCI was higher during the risk window than during the control window, with chronic pulmonary disease and mild liver disease observed in the risk window and peptic ulcer disease noted in the control window (**Table 1**).

There was no significantly increased risk of KD/MIS-C during the risk window compared with the control window within 60 days after COVID-19 vaccination (IRR, 0.53; 95% CI, 0.17–1.60). The secondary analysis, which considered the COVID-19 vaccination doses, showed the following results: first dose (IRR, 0.69; 95% CI, 0.16–3.08), second dose (IRR, 0.33; 95% CI, 0.07–1.45), third dose (IRR, 0.28; 95% CI, 0.01–10.34), and combined first and second doses (IRR, 0.46; 95% CI, 0.14–1.49). When analyzing both outcomes as separate diseases of interest, the results were not statistically significant for KD (IRR, 0.84; 95% CI, 0.18–3.87) and MIS-C (IRR, 0.19; 95% CI, 0.02–1.70). Based on the vaccine type, all individuals received BNT162b2. When all doses were consistently administered with the same type, the IRR (95% CI) was observed as 0.53 (0.17–1.60) (**Table 2**). Subgroup analyses stratified by sex and comorbidities also showed no significant increase in the risk of developing KD/MIS-C (**Supplementary Table 2**). Sensitivity analyses performed by adjusting the length of the risk window or redefining cases yielded results consistent with those of the primary analysis (**Supplementary Table 3**, **Table 3**). In **Table 3**, comorbidities that did not appear in the risk and control windows are not included (e.g., myocardial infarction, congestive heart failure, peripheral vascular disease, and cerebrovascular disease).

**Table 1.** Baseline characteristics of Kawasaki disease/multisystem inflammatory syndrome in children cases administered with coronavirus disease 2019 vaccines, stratified by exposure window

| Variables | Risk window | Control window | *P* |
|---|---|---|---|
| Total | 5 (100.0) | 7 (100.0) | |
| Age, yr | 14.8 ± 1.6 | 15.1 ± 1.5 | 0.719 |
| 0–4 | 0 (0.0) | 0 (0.0) | - |
| 5–11 | 0 (0.0) | 0 (0.0) | |
| 12–17 | 5 (100.0) | 7 (100.0) | |
| Sex | | | 0.276 |
| Male | 2 (40.0) | 5 (71.4) | |
| Female | 3 (60.0) | 2 (28.6) | |
| Health insurance type | | | - |
| NHI | 5 (100.0) | 7 (100.0) | |
| Medical aid | 0 (0.0) | 0 (0.0) | |
| Region of residence | | | 0.490 |
| Metropolitan | 5 (100.0) | 7 (100.0) | |
| Rural | 0 (0.0) | 0 (0.0) | |
| Comorbidities | | | |
| CCI | 0.6 ± 0.9 | 0.1 ± 0.4 | 0.331 |
| CCI < 3 | 5 (100.0) | 7 (100.0) | - |
| CCI ≥ 3 | 0 (0.0) | 0 (0.0) | |
| Chronic pulmonary disease | 2 (40.0) | 0 (0.0) | 0.067 |
| Rheumatic disease | 0 (0.0) | 0 (0.0) | - |
| Peptic ulcer disease | 0 (0.0) | 1 (14.3) | 0.377 |
| Mild liver disease | 1 (20.0) | 0 (0.0) | 0.217 |

Values are presented as number (%) or mean ± standard deviation.
NHI = National Health Insurance, CCI = Charlson comorbidity index.

**Table 2.** Risk of KD/MIS-C within 2 months of coronavirus disease 2019 vaccination, overall and stratified by dose, vaccine type

| Variables | No. of event | | Person-years | | Incidence rate | | IRR (95% CI) |
|---|---|---|---|---|---|---|---|
| | Risk | Control | Risk | Control | Risk | Control | |
| KD/MIS-C | | | | | | | |
| Overall | 5 | 7 | 5.33 | 3.94 | 0.94 | 1.78 | 0.53 (0.17–1.60) |
| Scheduled dose | | | | | | | |
| 1st dose | 2 | 7 | 1.63 | 3.94 | 1.23 | 1.78 | 0.69 (0.16–3.08) |
| 2nd dose | 2 | 7 | 3.28 | 3.78 | 0.61 | 1.85 | 0.33 (0.07–1.45) |
| 3rd dose | 1 | 1 | 0.42 | 0.12 | 2.36 | 8.49 | 0.28 (0.01–10.34) |
| 4th dose | 0 | 0 | - | - | - | - | - |
| 1st/2nd dose | 4 | 7 | 4.91 | 3.94 | 0.81 | 1.78 | 0.46 (0.14–1.49) |
| Vaccine type (1) | | | | | | | |
| Homologous | 5 | 7 | 5.33 | 3.94 | 0.94 | 1.78 | 0.53 (0.17–1.60) |
| Pfizer-BioNTech | 5 | 7 | 5.33 | 3.94 | 0.94 | 1.78 | 0.53 (0.17–1.60) |
| Novavax | 0 | 0 | - | - | - | - | - |
| Heterologous | 0 | 0 | - | - | - | - | - |
| Vaccine type (2) | | | | | | | |
| ≥ 1mRNA | 5 | 7 | 5.33 | 3.94 | 0.94 | 1.78 | 0.53 (0.17–1.60) |
| mRNA only | 5 | 7 | 5.33 | 3.94 | 0.94 | 1.78 | 0.53 (0.17–1.60) |
| mRNA + Others | 0 | 0 | - | - | - | - | - |
| Non-mRNA | 0 | 0 | - | - | - | - | - |
| KD[a] | | | | | | | |
| Overall | 4 | 3 | 2.24 | 1.42 | 1.78 | 2.12 | 0.84 (0.18–3.87) |
| 1st dose | 1 | 3 | 0.76 | 1.42 | 1.31 | 2.12 | 0.62 (0.06–6.17) |
| 2nd dose | 2 | 3 | 1.31 | 1.39 | 1.52 | 2.17 | 0.70 (0.12–3.98) |
| 3rd dose | 1 | 0 | 0.16 | 0.09 | 6.09 | - | - |
| 4th dose | 0 | 0 | - | - | - | - | - |
| 1st/2nd dose | 3 | 3 | 2.08 | 1.42 | 1.44 | 2.12 | 0.68 (0.13–3.46) |
| MIS-C[b] | | | | | | | |
| Overall | 1 | 4 | 3.38 | 2.60 | 0.30 | 1.54 | 0.19 (0.02–1.70) |
| 1st dose | 1 | 4 | 0.98 | 2.60 | 1.02 | 1.54 | 0.66 (0.09–5.07) |
| 2nd dose | 0 | 4 | 2.13 | 2.48 | - | 1.61 | - |
| 3rd dose | 0 | 1 | 0.26 | 0.02 | - | 40.58 | - |
| 4th dose | 0 | 0 | - | - | - | - | - |
| 1st/2nd dose | 1 | 4 | 3.12 | 2.60 | 0.32 | 1.54 | 0.21 (0.02–1.78) |

KD = Kawasaki disease, MIS-C = multisystem inflammatory syndrome in children, IRR = incidence rate ratio, CI = confidence interval, mRNA = messenger ribonucleic acid.
[a]Mean age 14.7 years; [b]Mean age 12.2 years.

## DISCUSSION

In this nationwide SCCS study, using a large-linked database that connects the COVID-19 immunization registry and claims data, no significant association was observed between KD/MIS-C within 60 days of COVID-19 vaccination. The results from the subgroup analyses stratified by sex and the presence of comorbidities were also consistent with those of the main analysis. Sensitivity analyses of the duration of the risk window and outcome definition yielded similar results, supporting the robustness of this study's findings.

Most studies on KD and MIS-C have predominantly focused on their association with SARS-CoV-2 infection,[23,24] while there is a paucity of evidence regarding the association between COVID-19 vaccines and the risk of KD/MIS-C. Although several case reports of the occurrence of KD/MIS-C following COVID-19 vaccination have been published,[10,11] suggesting a potential association, these reports alone do not establish a definitive association because of the low level of clinical evidence. In contrast, evidence suggests a protective effect of COVID-19 vaccines against these conditions. For instance, the US VAERS-based pharmacovigilance studies have observed lower reporting rates of KD/MIS-C in COVID-19 vaccinated individuals

**Table 3.** Risk of Kawasaki disease/multisystem inflammatory syndrome in children by varying the case definition

| Variables | No. of event | | Person-years | | Incidence rate | | IRR (95% CI) |
|---|---|---|---|---|---|---|---|
| | Risk | Control | Risk | Control | Risk | Control | |
| (1) Restricting to cases with a prescription record for IVIG | | | | | | | |
| Overall | 4 | 4 | 3.23 | 1.92 | 1.24 | 2.09 | 0.59 (0.15–2.31) |
| 1st dose | 1 | 4 | 1.00 | 1.92 | 1.00 | 2.09 | 0.48 (0.05–4.19) |
| 2nd dose | 2 | 4 | 1.97 | 1.88 | 1.02 | 2.12 | 0.48 (0.10–2.39) |
| 3rd dose | 1 | 0 | 0.26 | 0.09 | 3.84 | - | - |
| 4th dose | 0 | 0 | - | - | - | - | - |
| 1st/2nd dose | 3 | 4 | 2.97 | 1.92 | 1.01 | 2.09 | 0.48 (0.11–2.09) |
| (2) Restricting to cases with a prescription record for IVIG or other medications | | | | | | | |
| Overall | 8 | 9 | 7.46 | 5.93 | 1.07 | 1.52 | 0.71 (0.27–1.86) |
| 1st dose | 3 | 9 | 2.39 | 5.93 | 1.26 | 1.52 | 0.83 (0.23–2.93) |
| 2nd dose | 4 | 9 | 4.64 | 5.40 | 0.86 | 1.67 | 0.52 (0.16–1.63) |
| 3rd dose | 1 | 1 | 0.42 | 0.12 | 2.36 | 8.49 | 0.28 (0.01–10.34) |
| 4th dose | 0 | 0 | - | - | - | - | - |
| 1st/2nd dose | 7 | 9 | 7.03 | 5.93 | 1.00 | 1.52 | 0.66 (0.24–1.78) |
| (3) Excluding individuals with death recorded within 7 days from the date of outcome diagnosis | | | | | | | |
| Overall | 5 | 7 | 5.33 | 3.94 | 0.94 | 1.78 | 0.53 (0.17–1.60) |
| 1st dose | 2 | 7 | 1.63 | 3.94 | 1.23 | 1.78 | 0.69 (0.16–3.08) |
| 2nd dose | 2 | 7 | 3.28 | 3.78 | 0.61 | 1.85 | 0.33 (0.07–1.45) |
| 3rd dose | 1 | 1 | 0.42 | 0.12 | 2.36 | 8.49 | 0.28 (0.01–10.34) |
| 4th dose | 0 | 0 | - | - | - | - | - |
| 1st/2nd dose | 4 | 7 | 4.91 | 3.94 | 0.81 | 1.78 | 0.46 (0.14–1.49) |
| (4) Excluding individuals with COVID-19 within 60 days before vaccination | | | | | | | |
| Overall | 5 | 7 | 5.33 | 3.94 | 0.94 | 1.78 | 0.53 (0.17–1.60) |
| 1st dose | 2 | 7 | 1.63 | 3.94 | 1.23 | 1.78 | 0.69 (0.16–3.08) |
| 2nd dose | 2 | 7 | 3.28 | 3.78 | 0.61 | 1.85 | 0.33 (0.07–1.45) |
| 3rd dose | 1 | 1 | 0.42 | 0.12 | 2.36 | 8.49 | 0.28 (0.01–10.34) |
| 4th dose | 0 | 0 | - | - | - | - | - |
| 1st/2nd dose | 4 | 7 | 4.91 | 3.94 | 0.81 | 1.78 | 0.46 (0.14–1.49) |
| (5) Excluding individuals with COVID-19 within 60 days before outcome diagnosis | | | | | | | |
| Overall | 5 | 7 | 3.30 | 3.22 | 1.51 | 2.17 | 0.70 (0.24–2.05) |
| 1st dose | 2 | 7 | 1.00 | 3.22 | 1.99 | 2.17 | 0.92 (0.23–3.73) |
| 2nd dose | 2 | 7 | 1.70 | 3.07 | 1.01 | 2.28 | 0.45 (0.11–1.89) |
| 3rd dose | 1 | 1 | 0.33 | 0.12 | 3.04 | 8.49 | 0.36 (0.01–12.82) |
| 4th dose | 0 | 0 | - | - | - | - | - |
| 1st/2nd dose | 4 | 7 | 2.98 | 3.22 | 1.34 | 2.17 | 0.62 (0.20–1.93) |
| (6) Excluding individuals with diagnosis of bacterial sepsis, toxic shock syndrome, or infectious myocarditis on the same date of outcome diagnosis | | | | | | | |
| Overall | 5 | 7 | 5.33 | 3.94 | 0.94 | 1.78 | 0.53 (0.17–1.60) |
| 1st dose | 2 | 7 | 1.63 | 3.94 | 1.23 | 1.78 | 0.69 (0.16–3.08) |
| 2nd dose | 2 | 7 | 3.28 | 3.78 | 0.61 | 1.85 | 0.33 (0.07–1.45) |
| 3rd dose | 1 | 1 | 0.42 | 0.12 | 2.36 | 8.49 | 0.28 (0.01–10.34) |
| 4th dose | 0 | 0 | - | - | - | - | - |
| 1st/2nd dose | 4 | 7 | 4.91 | 3.94 | 0.81 | 1.78 | 0.46 (0.14–1.49) |

IRR = incidence rate ratio, CI = confidence interval, IVIG = intravenous immunoglobulin, COVID-19 = coronavirus disease 2019.

compared to unvaccinated individuals.[11] Additionally, another study has shown that COVID-19 vaccination reduces the incidence of MIS-C by reducing the risk of SARS-CoV-2 infection.[25] Our study advances these prior investigations by differentiating between the impacts of COVID-19 vaccination and infection, enabling us to assess their association with KD/MIS-C through SCCS analysis. The strength of our study design lies in the ability of the SCCS to use within-person comparisons, thereby effectively controlling for time-invariant factors such as genetics, which is another key component in the occurrence of KD/MIS-C.[26]

KD has been associated not only with viral infections but also with vaccination. A few previous studies have established the association between KD and various vaccines, such as those for chickenpox, influenza, pneumococcus, or hepatitis A or B.[7] Specifically, a US-based

study reported that vaccination reduced the incidence of KD,[27] while a UK study found no effect of vaccines such as the pneumococcal conjugate vaccine and meningococcal B vaccine, on the incidence of KD.[28] Considering that research results vary across countries, even for non-COVID-19 vaccines, and there is limited research specifically addressing the relationship between COVID-19 vaccines and KD occurrence. Therefore, it is currently deemed inappropriate to establish causation based on the available evidence. Further research is required to investigate the association between COVID-19 vaccinations and KD/MIS-C.

In addition to MIS-C, cases of multisystem inflammatory syndrome in adults (MIS-A) have also been reported during COVID-19 pandemic. Majority of the case reports were those following confirmed or suspected COVID-19 infection,[29,30] and cases reported in association with SARS-CoV-2 vaccines are rare. Similar to its counterpart in children (MIS-C), MIS-A involves widespread inflammation affecting multiple organ systems.[31] MIS-A has been suggested to have a stronger temporal association with COVID-19 infection compared to MIS-C,[31] leading to the hypothesis that interventions preventing COVID-19 might also prevent MIS-A. As such, it would be valuable to conduct follow-up studies that confirm the protective effect of vaccination against MIS-A.

This study is significant because it represents a nationwide investigation of the association between COVID-19 vaccination and KD/MIS-C using the SCCS analysis. The strength of this study is that it reduces the bias inherent in cohort studies. By comparing each individual to their own control, the SCCS method minimizes confounding factors associated with individual characteristics, thereby enhancing the reliability of the findings.[16] Furthermore, SCCS analysis can be effectively employed for studies involving small sample sizes, making it well-suited for studies of extremely rare adverse events such as KD/MIS-C. This enables a precise examination of the temporal association between specific events and exposures, and demonstrates the ability of the SCCS analysis method to generate statistically significant results.

Our study has several limitations. First, despite utilizing a nationwide, large-linked database encompassing all vaccinated children in South Korea, there were challenges in identifying sufficient cases of KD/MIS-C. In particular, vaccination of children aged < 5, the most prevalent age group for KD, began on February 13, 2023, resulting in a short exposure assessment period. This delayed vaccination campaign for this age group may have led to an underestimation of cases diagnosed with KD/MIS-C after vaccination. Second, the SCCS analysis did not account for time-varying confounders, and residual confounding may exist owing to the observational nature of the study. Third, although we identified KD and MIS-C separately, outcome misclassification may have occurred from relying solely on the diagnosis codes and drug prescriptions in the claims data. Specifically, the fact that the mean age of the population with KD in this study was higher than that of MIS-C suggests that they may have not been separated correctly. Fourth, since this study was conducted using claims data, it was unable to incorporate clinical criteria (such as fever symptoms, C-reactive protein levels, and platelet count) or laboratory criteria for prior COVID-19 infection (such as polymerase chain reaction results and serology test results) in the definition of KD/MIS-C. Lastly, this study only used data from South Korea, which may limit its generalizability to other ethnicities. Nevertheless, South Korea is known to have a high prevalence of KD, exceeding that of the US and Europe by 10–30 times, and ranking second only to Japan globally.[32] Hence, the findings of this study can be deemed moderately representative.

In summary, this study offers perspectives on the association between COVID-19 vaccination and KD/MIS-C in children, addressing a gap in existing research in which knowledge remains limited. Despite the potential impact of effectively regulating systemic inflammatory responses during the acute phase on mortality rates, the exact biological mechanisms of KD/MIS-C remain unclear. With the rising incidence of these conditions, the importance of a thorough understanding and monitoring of KD/MIS-C has become increasingly evident. Given the low absolute prevalence of these diseases, it would be beneficial to conduct follow-up studies on the same topic after we have more data accumulated on COVID-19 vaccination in pediatric population, especially infants and young children less than 5 years.

## SUPPLEMENTARY MATERIALS

### Supplementary Table 1

Diagnostic and procedure codes, prescription codes for KD and MIS-C for case definition

### Supplementary Table 2

Risk of KD/MIS-C within 60 days of coronavirus disease 2019 vaccination, stratified by sex, comorbidities

### Supplementary Table 3

Risk of Kawasaki disease/multisystem inflammatory syndrome in children following coronavirus disease 2019 vaccination that varied the risk window length

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